

Chapter 1: Cellular Biology

The cell is defined as the basic structural and functional unit of all living organisms. Every organism is composed of one or more cells. The cell theory is a fundamental principle of biology that states that all living things are composed of cells and that new cells are produced from existing cells.

Chapter 2: Photosynthesis

Photosynthesis is defined as the process by which green plants transform light energy into chemical energy. Photosynthesis occurs in the chloroplast, which contains the green pigment chlorophyll. Chlorophyll is the pigment that absorbs light energy. The overall equation for photosynthesis shows that carbon dioxide and water, in the presence of light, produce glucose and release oxygen.

Chapter 3: Cellular Respiration

Cellular respiration is the process by which cells break down glucose to produce ATP. Cellular respiration consists of three main stages. Glycolysis occurs in the cytoplasm and is the first stage, during which glucose is broken down into two molecules of pyruvate. The Krebs cycle occurs in the mitochondrial matrix. The electron transport chain is the final stage and occurs on the inner mitochondrial membrane.